

Statement of Work

Study Coordinating Center for the Lung Health Cohort

1. Background

Developing a robust approach to lung disease prevention requires a broad definition of lung health. While the respiratory research community has been limited to defining lung health as the absence of disease, the cardiovascular (CV) community has explicitly delineated a continuum ranging from ideal health to impaired health to define disease. This continuum allows for the identification of pre-disease states as well as modifiable risk factors that can be used to monitor and mitigate the worsening of CV health over time. There has been growing interest to develop a similar paradigm for lung disease, in which “Lung Health” would be defined and understood as a foundation for developing preventive health measures by targeting risk factors that could be used to slow progression toward, mitigate, or avoid lung disease.

The Lung Health Cohort (LHC) Study is motivated by the importance of examining a large cohort of healthy young adults focused on the respiratory system in order to define the constructs of ‘ideal’ versus ‘impaired’ lung health. Previous studies of subclinical lung disease have examined healthy controls in disease-focused studies or have used limited information about respiratory symptoms, physiology or imaging derived from cohort studies that were not focused on respiratory health. Life course studies of lung health are necessary to describe how an individual progresses from ideal lung health to an intermediate phenotype of impaired lung health and ultimately to chronic lung disease. Therefore, the current conceptual paradigm of the LHC assumes that there are two fundamental variables that define good or impaired lung health: (1) ‘reserve’ or peak lung function in young adulthood and (2) ‘susceptibility’ to impaired lung health or rate of decline in lung function from young adulthood through middle and old age. The LHC will focus on modifiable exposures and clinical and behavioral risk factors in members of the millennial generation to identify associations with impaired lung health reserve and on biomarkers of susceptibility to future lung disease. Identifying the factors associated with reserve and susceptibility will enhance the depiction of the continuum of lung health and will facilitate designing novel prevention strategies.

The LHC is currently funded by a single NHLBI cooperative agreement (U01), first awarded in 2019 and expected to end in 2026, which supports the LHC study coordinating center. The cooperative agreement supports study enrollment, data coordination, human subjects regulation, imaging oversight, environmental surveillance, and other functions. The study

leverages the existing infrastructure of the American Lung Association (ALA) Airways Clinical Research Centers (ACRC) network to enroll 4000 participants between ages 25 and 35 years old at clinical sites in 17 metropolitan regions across the USA. Study procedures for the baseline visit include physical assessment, spirometry, chest computed tomography (CT) scan, accelerometry, and collection of nasal epithelial lining fluid, nasal epithelial cells, blood, and urine. Additionally, medically related participant questionnaires are being recorded such as sociodemographic characteristics, home address histories and exposures, work history and exposures, medical histories, lung health, and health behaviors and activity. While baseline visits are ongoing, the full cohort of 4000 participants is expected to have been enrolled and complete a baseline visit by fall of 2026 under the current cooperative agreement.

2. Objectives

This requirement will be for a study coordinating center to support the continuation of LHC. The contractor shall complete a second in-person visit of the full LHC cohort, to be completed approximately five years after the baseline visit, as well as other data to be collected remotely, all to be conducted through the ACRC network. Data and biospecimens shall be assessed for quality, stored, analyzed, and distributed as appropriate. Finally, the study shall be managed in a manner consistent with NIH and NHLBI policies with oversight from a single Institutional Review Board (IRB) and an NHLBI-managed Observational Study Monitoring Board (OSMB). The objective of this contract is to exploit the unique characteristics of the LHC to clarify how potential factors might alter ideal lung health and increase risk for future diseases such as chronic obstructive pulmonary disease.

The objective remains to support continuation of the Lung Health Cohort (LHC) Study; complete second follow-up visits; manage data/biospecimens; perform analyses; and maintain compliance with NIH/NHLBI policies including this SOW's Section 508 requirements.

3. Task Areas

There are two main task areas for this acquisition.

Task 1.0: Program Management: The contractor shall facilitate the management of study personnel, data, and biospecimens to enable achievement of study goals.

The contractor shall:

1. Organize residential, school and occupational address histories, which will be used to enable geocoding. Geocodes will be used to link relevant exposure data to each participant, including such exposures as elemental carbon, organic carbon, nitrogen dioxide, atmospheric particulate matter, ozone, proximity to greenspace, census-tract level indicators of neighborhood socioeconomic position, and local daily temperature.
2. Evaluate the quality of CT scans collected during study visits. Analyze data of images collected during baseline and follow-up study visits.
3. Analyze LHC data to address the major scientific goals of the study, generate study manuscripts, and publish in peer-reviewed journals. Overall, this work is expected to result in at least one peer-reviewed publication per year throughout the contract period. Provide statistical analytical support to study investigators within limits of resources provided by the Data Coordinating Center (DCC).
4. Perform statistical analyses and power considerations.
5. Communicate with a designated and approved single IRB to obtain approval for the main study protocol and ancillary studies, as well as to report adverse events and other deviations as specified by NIH policies.
6. Schedule, plan, and organize annual meetings and ad hoc meetings as directed of an Observational Study Monitoring Board (OSMB), consisting of between 4 and 7 members. Meetings will be virtual unless otherwise specified by the COR. Provide compensation to OSMB members. Provide expedited reporting to the OSMB of serious unexpected adverse events and unanticipated problems as defined and required by NHLBI policies. Provide annual reports to the OSMB of study progress, including enrollments and visit completions, adverse events, protocol deviations, data quality and timeliness, presentations, publications, and major study findings. (The NHLBI Adverse Event (AE) and Unanticipated Problem (UP) Reporting Policy describes the procedures for identifying, monitoring, and reporting adverse events and unanticipated problems (<http://www.nhlbi.nih.gov/funding/policies/adverse.htm>).
7. Oversee general study management using a Data Coordinating Center, and Imaging Core, an Environmental Core, and other study centers and cores, as approved by NHLBI.
8. Establish and manage committees as needed for study governance. These may be organized through the ACRC network.

9. Travel to and register for domestic and international scientific meetings to discuss and present study data. **[NOTE TO OFFERORS: For cost proposal purposes assume one trip per year for \$4,000 per person. The offeror should propose the appropriate no. of people for their site (the Study Coordinating Center) and the participating sites/centers should include the trip in their respective budgets.]**
10. Travel to conduct study site visits and training, as needed. **[NOTE TO OFFERORS: For cost proposal purposes assume one trip per year for one night.]**
11. Oversee a data repository with quality control of study data. The data repository shall implement data science methods, standards, and defined metadata that are consistent with NHLBI recommendations.
12. Provide quality control checks of data submitted from the clinical centers, and work with the clinical centers to correct errors. Collect and store in the data repository newly collected data, including data from study Visits, long-term follow up contacts, environmental monitoring, and biospecimen analyses. Incorporate ancillary study data into the contract study data according to NIH and NHLBI policies. Provide appropriate data quality control.
13. Maintain a biospecimen repository for Lung Health Cohort biospecimens. This repository will store biospecimens under conditions established by protocol to minimize degradation over time and will employ backup systems to ensure preservation of biospecimens in the event of power losses. Distribute biospecimens to investigators for approved ancillary studies.
14. Transport biospecimens to an NHLBI-approved biorepository upon study conclusion.
15. Comply with NHLBI Data Management and Sharing Policies, including the NHLBI Supplement to the NIH Policy for Data Management and Sharing (this can be found at <https://www.nhlbi.nih.gov/grants-and-training/policies-and-guidelines/nhlbi-policy-for-data-sharing>). Compliance will involve regular deposition of all available cleaned study data and ancillary study data to an NHLBI-approved scientific data repository such as NHLBI's BioData Catalyst. The schedule of data deposition will be mutually agreed upon by the contractor and NHLBI, but is expected to be approximately once per year. Data sets shall include documentation of data formats and study procedures.
16. Provide regular progress reports to NHLBI that include enrollment and retention rates of Lung Health Cohort participants, in person and telephonic visit completion rates, data completeness, data quality measures, numbers of manuscripts and

publications, numbers of funded ancillary studies, and number of Early Stage Investigators funded to conduct ancillary studies.

17. Maintain the study website that is at the ALA. The website shall be updated and be 508-compliant within 6 months of contract award. Maintain and update the website throughout the contract period. The study website will include:
 - a. A description of the LHC study.
 - b. A description of available data, how to request data, and how to propose ancillary studies.
 - c. A description of available biospecimens and the procedures for requesting and obtaining approval for obtaining biospecimens.
 - d. Links to any current study policies, as appropriate.
18. Ensure the website meets Section 508 Standards and WCAG 2.1 A/AA, including support for screen readers, keyboard navigation, proper HTML structure, alt-text for images, compliant color contrast, and fully accessible downloadable documents, and shall provide a VPAT/ACR at launch.
19. The Contractor shall perform all following Section 508 compliance requirements for any Information and Communication Technology (ICT) products, including but not limited to:
 - a. Study website
 - b. Digital reports, PDFs, Word documents, PowerPoints
 - c. Data dashboards and repositories
 - d. Participant-facing tools and digital forms
 - e. Electronic communications to participants
 - f. Training materials
 - g. Any software or web-based application
20. Comply with the NIH Public Access Policy To that end, the Contractor shall do so for any manuscript that arises from any direct funding from the contract. For more information, see NOTOD-08-033 (<http://grants.nih.gov/grants/guide/notice-files/NOT-OD-08033.html>) and the Public Access website (<https://publicaccess.nih.gov/>).
21. Comply with all other NHLBI and NIH policies.
22. If directed by the NHLBI, provide training and support for transitioning out study operations to a successor contractor, in accordance with FAR 52.237-3, Continuity of Services (Jan 1991). To that end, work closely with the new entity to:

- a. Provide a smooth and seamless transition of responsibilities to the successor.
- b. Help the successor obtain IRB approval with appropriate consideration for conditions that have been established and existing in the current contract, including informed consent conditions agreed to by the participants.
- c. Provide smooth transfer of all biological samples and data, including data files, scans, and images, and assurances of confidentiality and security of all data as necessary.
- d. Transfer all existing protocols, relevant documents, minutes, and reports to the successor as necessary.
- e. Provide transition of the internal, public access, and participant websites.

Task 2.0: Follow-Up Exam and Data Collection: The contractor shall use the ACRC network to complete a five-year follow-up exam and remote follow-up data collection of the Lung Health Cohort.

The contractor shall:

1. Complete an in-clinic follow-up exam of up to 4,000 Lung Health Cohort participants. Visits will be scheduled approximately five years after each participant's baseline visit. This visit will be conducted according to an NHLBI-approved protocol and will include the key data listed below. Some subjects may refuse some aspects of the visit but may still be considered to have a complete visit. Study visits shall be conducted through the ACRC network and may include (as defined by NHLBI-approved protocol):
 - a. Physical assessment (e.g. height, weight, resting heart rate, blood pressure)
 - b. Spirometry
 - c. Pregnancy testing in participants of child-bearing potential
 - d. Chest CT scan (to be transferred to a central Imaging Core)
 - e. Questionnaires (including but not limited to: contact information, address history, demographics, lung health, substance use, environmental and occupational exposures, medication use, medical history, activity, sleep, social elements, nutrition)
 - f. Nasosorption strip collection
 - g. Nasal epithelial curettage
 - h. Phlebotomy
 - i. Urine collection

2. Process biospecimens collected at baseline exam (blood, nasal epithelial cells, and urine.) Each sample will be labeled with a unique bar code that includes no health information but can be linked to the participant ID.
3. Ship biospecimens to ACRC study biorepository for long-term storage. Biospecimens collected under this contract will be the property of the United States Government.
4. Collect data remotely on the full Cohort. Remote data collection will be according to an NHLBI-approved protocol and may include (as defined by NHLBI-approved protocol):
 - a. Home spirometry
 - b. Actigraphy monitoring
 - c. Activity log
 - d. Vaping log
5. Collect remote follow-up data on up to 4000 Lung Health Cohort participants every 3 to 6 months (up to four times a year) during which participants may be asked to (as defined by NHLBI-approved protocol):
 - a. Update contact information
 - b. Complete additional assessments related to topics from the in-person visit
 - c. Complete additional assessments related to health events
 - d. Schedule a coached at-home spirometry session, to be completed at a later date.
6. Contact study participants via telephone or internet to collect additional information if deemed necessary by the approved study protocol or by NHLBI.
7. For study close-out complete raw data QA/QC process (including imaging results), and share comprehensive dataset(s) from all recruited participants. Complete any remaining exam related communications such as participant results reports.
8. Remote Data Collection Tools (Participant-facing ICT):
 - a. Accessible Remote Tools - Every participant-facing ICT element must:
 - i. Meet WCAG 2.1 A/AA
 - ii. Provide accessible form fields, labels, navigation
 - iii. Support screen readers
 - iv. Provide captioning for multimedia instructions
 - v. Provide accessible alternatives to non-accessible devices/interfaces
 - b. Accessible Questionnaires & Logs - All electronic questionnaires (baseline, follow-up, environmental, occupational, etc.) must:

- i. Be accessible via screen reader
 - ii. Provide properly labeled fields
 - iii. Allow keyboard-only navigation
 - iv. Avoid inaccessible modal windows
 - v. Use clear error messaging readable by assistive tech
- c. Accessible Participant Communication - Digital communications must:
 - i. Use accessible templates
 - ii. Contain properly structured headings
 - iii. Avoid images without alt-text
 - iv. Meet color contrast requirements

4. Section 508 Acceptance Criteria

Deliverables will be accepted only if:

- a. 100% of WCAG Level A criteria are met
- b. 98–100% of WCAG Level AA criteria are met
- c. No critical blockers exist
- d. ACR/VPAT is provided
- e. All PDFs pass accessibility checkers
- f. ICT delivers functional parity for users with disabilities

NHLBI reserves the right to reject deliverables that fail any accessibility tests.

5. Section 508 Review and Oversight

NHLBI will perform:

- a. Independent accessibility testing
- b. Periodic 508 reviews
- c. Validation of VPAT/ACRs
- d. Verification of remediation

Contractor must support all audits and provide additional documentation as requested.

6. Additional Requirements

- a. Contractor shall maintain all accessibility records for the contract period.
- b. Contractor shall ensure subcontractors meet all 508 requirements.
- c. Provide yearly VPAT/ACRs

- d. All ICT updates must undergo accessibility testing before release.